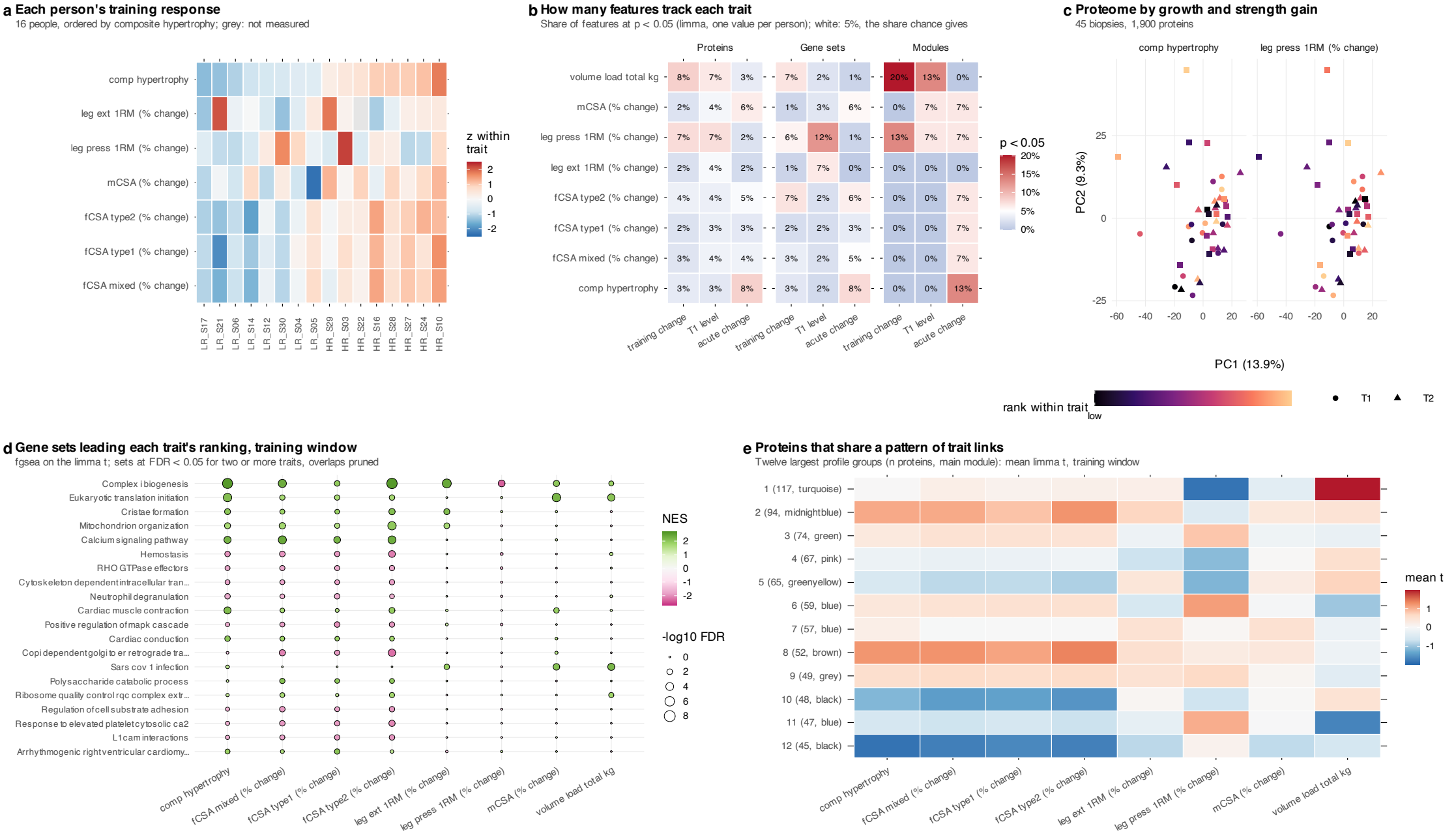
a Each person's training response
16 people, ordered by composite hypertrophy; grey: not measured
comp hypertrophy
leg ext 1RM (% change)
leg press 1RM (% change)
mCSA (% change)
fCSA type2 (% change)
fCSA type1 (% change)
fCSA mixed (% change)
LR_S17 LR_S21 LR_S06 LR_S14 LR_S12 LR_S30 LR_S04 LR_S05 HR_S29 HR_S03 HR_S22 HR_S16 HR_S28 HR_S27 HR_S24 HR_S10
z within trait
2 1 0 -1 -2
b How many features track each trait
Share of features at p < 0.05 (limma, one value per person); white: 5%, the share chance gives
Proteins
Gene sets
Modules
volume load total kg 8% 7% 3% 7% 2% 1% 20% 13% 0%
mCSA (% change) 2% 4% 6% 1% 3% 6% 0% 7% 7%
leg press 1RM (% change) 7% 7% 2% 6% 12% 1% 13% 7% 7%
leg ext 1RM (% change) 2% 4% 2% 1% 7% 0% 0% 0% 0%
fCSA type2 (% change) 4% 4% 5% 7% 2% 6% 0% 0% 7%
fCSA type1 (% change) 2% 3% 3% 2% 2% 3% 0% 0% 7%
fCSA mixed (% change) 3% 4% 4% 3% 2% 5% 0% 0% 7%
comp hypertrophy 3% 3% 8% 3% 2% 8% 0% 0% 13%
training change
T1 level
acute change
p < 0.05
20% 15% 10% 5% 0%
c Proteome by growth and strength gain
45 biopsies, 1,900 proteins
comp hypertrophy
leg press 1RM (% change)
PC2 (9.3%)
PC1 (13.9%)
rank within trait
low
T1
T2
d Gene sets leading each trait's ranking, training window
fgsea on the limma t; sets at FDR < 0.05 for two or more traits, overlaps pruned
Complex i biogenesis
Eukaryotic translation initiation
Cristae formation
Mitochondrion organization
Calcium signaling pathway
Hemostasis
RHO GTPase effectors
Cytoskeleton dependent intracellular tran...
Neutrophil degranulation
Cardiac muscle contraction
Positive regulation of mapk cascade
Cardiac conduction
Copi dependent golgi to er retrograde tra...
Sars cov 1 infection
Polysaccharide catabolic process
Ribosome quality control rqc complex extr...
Regulation of cell substrate adhesion
Response to elevated platelet cytosolic ca2
L1cam interactions
Arrhythmogenic right ventricular cardiomy...
comp hypertrophy
fCSA mixed (% change)
fCSA type1 (% change)
fCSA type2 (% change)
leg ext 1RM (% change)
leg press 1RM (% change)
mCSA (% change)
volume load total kg
NES
2 1 0 -1 -2
-log10 FDR
0 2 4 6 8
e Proteins that share a pattern of trait links
Twelve largest profile groups (n proteins, main module): mean limma t, training window
1 (117, turquoise)
2 (94, midnightblue)
3 (74, green)
4 (67, pink)
5 (65, greenyellow)
6 (59, blue)
7 (57, blue)
8 (52, brown)
9 (49, grey)
10 (48, black)
11 (47, blue)
12 (45, black)
comp hypertrophy
fCSA mixed (% change)
fCSA type1 (% change)
fCSA type2 (% change)
leg ext 1RM (% change)
leg press 1RM (% change)
mCSA (% change)
volume load total kg
mean t
1 0 -1